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Smart Textile Manufacturing Quality Inspection and Production Optimization Platform

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

CSA1016

Software Engineering

to the award of the degree of

BACHELOR OF ENGINEERING

IN

INFORMATION TECHNOLOGY

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MAY 2026

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Executive Summary :

Textile manufacturing is a high-volume industry in which product quality and production efficiency have a direct effect on cost, customer satisfaction, and delivery performance. Despite this, many manufacturing units continue to depend on manual visual inspection and separate production records. These practices can lead to missed defects, inconsistent inspection results, delayed corrective action, material wastage, and difficulty in identifying the causes of recurring quality problems.

This report proposes an integrated Smart Textile Manufacturing Quality Inspection and Production Optimization Platform. The proposed system combines computer vision and machine learning for automated defect detection, IoT-based production monitoring, centralized data storage, real-time dashboards, automated alerts, root-cause analysis, digital reporting, and data-driven production recommendations.

The solution is intended to support human decision-making rather than simply replace existing personnel. Quality inspectors can use the system to identify and verify defects, production managers can monitor performance and bottlenecks, and maintenance teams can use historical patterns to investigate recurring problems. The report also recommends Agile Scrum as the development methodology because the project involves evolving requirements, AI model development, data integration, and continuous feedback from users.

1. Introduction :

1.1 Industry Background

The textile manufacturing industry includes processes such as weaving, knitting, dyeing, stitching, and finishing. Each stage can introduce quality problems that affect the appearance, strength, consistency, or usability of the finished product. As manufacturers operate under increasing pressure to reduce costs and deliver consistent quality, efficient inspection and production monitoring have become increasingly important.

Traditional inspection methods rely heavily on the experience and concentration of human inspectors. While human expertise remains valuable, manual inspection becomes difficult to maintain at a consistent level when production volumes are high or when employees work for long periods. Digital technologies therefore provide an opportunity to improve the speed, consistency, and traceability of quality control.

1.2 Problem Context

The problem considered in this case study is the lack of an integrated system for quality inspection and production optimization in textile manufacturing. Fabric defects may be detected late, while information about machines, production speed, downtime, shifts, and raw material batches may be stored separately. As a result, the factory may identify a quality problem without being able to quickly determine the operational conditions that caused it.

1.3 Purpose of the Report

The purpose of this report is to analyze the identified textile manufacturing problem and propose a realistic software engineering solution. The report considers the stakeholders involved, the weaknesses of the current approach, the proposed system and its requirements, the way the system solves the problem, the development methodology, and the expected benefits and limitations.

2. Understanding the Problem Statement :

2.1 Problem Analysis

The main issue is that textile quality inspection and production decision-making are not sufficiently integrated. Manual inspection can be slow and inconsistent, and defects may not be discovered until a later stage of production. When a defect is discovered late, the factory may already have produced a large quantity of affected material. This increases the possibility of waste, rework, and customer dissatisfaction.

A second issue is the absence of a centralized relationship between production conditions and quality results. If a defect repeatedly occurs, managers may need to compare information from several sources, such as machine records, operator information, shift details, and material records. This makes root-cause analysis slower and encourages reactive rather than proactive decision-making.

2.2 Objectives

- 1.Design an automated fabric defect detection module using image processing and machine learning.
- 2.Develop a real-time production monitoring dashboard that tracks machine performance, output rate, and downtime.
- 3.Classify defects by type, severity, and location on the fabric roll.
- 4.Correlate defect patterns with machines, shifts, production conditions, and raw material batches to support root-cause analysis.
- 5.Develop an optimization component that recommends improvements to machine speed and production scheduling.
- 6.Generate automated alerts when defect rates or other important indicators exceed acceptable thresholds.
- 7.Replace disconnected manual reporting with a centralized digital reporting system.
- 8.Ensure that the platform can be adapted to different textile processes, including weaving, knitting, and dyeing.

2.3 Scope

The proposed system focuses on the following activities:

- Automated fabric image inspection and defect detection.
- Defect classification and severity assessment.
- Monitoring of machine and production data.
- Real-time dashboards and alerts.
- Historical analysis and root-cause investigation.

- Digital reporting and production recommendations.

The initial system is not intended to replace all human decision-making or provide fully autonomous control of factory machinery. Recommendations generated by the platform should be reviewed and approved by authorized personnel, particularly when changes could affect safety or production quality.

3. Stakeholder and Challenge Analysis :

3.1 Stakeholders

Stakeholder	Role	Main Needs
Factory Management	Makes strategic and financial decisions	Reliable performance, quality, cost, and productivity information
Production Manager	Plans and supervises production	Machine utilization, output, downtime, and scheduling information
Quality Inspector	Checks product quality	Accurate defect information and ability to verify AI results
Machine Operator	Operates production machinery	Clear alerts and understandable machine information
Maintenance Team	Maintains equipment	Downtime trends and early indicators of machine problems
IT Administrator	Maintains the system	Security, availability, integration, and maintainability
Customers	Receive manufactured products	Consistent quality and timely delivery

3.2 Stakeholder Challenges

Different stakeholders experience different effects from the current problem. For example, management requires reliable information for decisions, while quality

inspectors need tools that reduce repetitive manual work without removing their ability to apply professional judgment. Production and maintenance teams need information that is timely enough to support action rather than merely describe problems after they have occurred.

4. Analysis of the Current Approach :

4.1 Existing Process

In a typical manual process, fabric is produced and inspected by personnel during or after production. Defects may be recorded manually, while machine performance and production information may be stored separately. If a problem is identified, supervisors may need to investigate several records before deciding what action to take.

Production → Manual Inspection → Defect Recording → Supervisor Review → Investigation → Separate Reporting → Corrective Action

4.2 Strengths of the Current Approach

- Experienced inspectors can recognize unusual defects and use human judgment.
- The process can operate without major investment in advanced technology.
- Human staff can consider contextual information that may not be represented in a dataset.
- The approach may be practical for smaller production volumes.

4.3 Weaknesses and System Gaps

Weakness / Gap	Effect on Manufacturing
Manual and subjective inspection	Inspection results may vary between people and situations.
Human fatigue	missed defects may increase during long shifts.

Late detection	More material may be produced before the problem is identified.
Disconnected data	It is difficult to relate quality problems to machine and production conditions.
Manual reporting	Analysis and decision-making take more time.
Reactive maintenance	Unexpected downtime can reduce production capacity.
Experience-based scheduling	Machines may not be used in the most efficient way.

5. How the Proposed System Solves the Problem :

The proposed solution changes the manufacturing process from a mainly reactive approach to a more continuous and data-driven process. The system does this by collecting information during production, detecting problems early, connecting quality results with operational data, and presenting useful information to the people responsible for taking action.

5.1 Problem-to-Solution Relationship

Current Problem	Proposed Response
Manual visual inspection is time-consuming.	Use camera-based inspection supported by computer vision and machine learning.
Defects may be missed or detected late.	Monitor fabric continuously and provide early alerts.
Inspection results can be subjective.	Use standardized defect categories and model-based classification.
Quality and machine data are disconnected.	Store related information in a centralized platform.

Root causes are difficult to identify.	Compare defects with machine, shift, speed, and material information.
Production planning is reactive.	Use historical and real-time data to support scheduling recommendations.
Reports are maintained manually.	Generate centralized digital quality and productivity reports.

5.2 Step-by-Step Solution

- Cameras capture images of the fabric as it moves through the production process.
- The system preprocesses the images to make them suitable for analysis.
- A machine-learning model identifies supported defects and estimates their classification.
- The system records the defect type, severity, location, time, machine, shift, and relevant production information.
- IoT sensors and machine interfaces provide data about speed, output, downtime, temperature, vibration, or other available parameters.
- The analytics module compares defect patterns with production conditions to identify possible relationships.
- The alert module notifies responsible personnel when defect rates or machine conditions exceed defined limits.
- The system provides recommendations that may help managers adjust production schedules, machine settings, or maintenance priorities.
- Reports are generated from the same centralized data source, improving traceability and reducing dependence on disconnected spreadsheets.

5.3 Example of Root-Cause Analysis

Assume that the system observes a repeated increase in broken-thread defects. The platform can compare the defect records with operating conditions and discover that the

problem occurs mainly on one machine, at high speed, during a particular shift, or with a particular raw material batch.

This does not automatically prove that one factor is the cause. However, it gives the production and maintenance teams a focused starting point for investigation. This is more useful than simply recording that a defect occurred because it supports evidence-based corrective action.

5.4 Example of Production Optimization

If historical records show that a machine produces a higher defect rate when operating above a certain speed, the system may recommend a lower operating range for that machine. Similarly, if a machine has frequent downtime, urgent orders may be assigned to another machine with better availability. The platform therefore supports managers with evidence while leaving final operational decisions to authorized personnel.

6. Proposed Software Solution :

6.1 System Overview

The proposed platform consists of several connected components. Cameras and sensors collect information from the production environment. The data is processed by the defect detection and monitoring services, stored in a centralized database, and then presented through dashboards, alerts, reports, and analytical features.

Production Data → Processing and Analysis → Centralized Storage → Dashboard, Alerts, Reports and Recommendations



6.2 Major Functional Modules

Module	Purpose
User and Role Management	Controls access to the system according to user responsibilities.
Image Acquisition and Preprocessing	Receives and prepares fabric images for analysis.
Defect Detection	Identifies visible defects using computer vision and machine learning.
Defect Classification	Records defect type, severity, location, and confidence information.
Production Monitoring	Tracks machine status, speed, output, and downtime.
Dashboard	Presents quality and production indicators in a usable form.
Alerts	Notifies users when important thresholds are exceeded.
Root-Cause Analysis	Examines relationships between defects and production conditions.
Reporting	Produces digital quality, production, and machine reports.
Optimizatin Support	Provides recommendations for scheduling and production improvement.

6.3 Functional Requirements

- **User Login:** The system should allow authorized users to log in securely.
- **Defect Detection:** The system should automatically detect fabric defects using images and machine learning.
- **Defect Classification:** The system should classify defects according to their type, severity, and location.
- **Production Monitoring:** The system should monitor machine performance, production speed, output, and downtime.
- **Data Collection:** The system should collect relevant data from machines, cameras, and IoT sensors.
- **Alert Generation:** The system should send alerts when defect rates or machine problems exceed acceptable limits.
- **Root-Cause Analysis:** The system should help identify possible causes of defects by analyzing machine, shift, and material data.
- **Production Optimization:** The system should provide recommendations for improving machine utilization, production scheduling, and efficiency.
- **Dashboard:** The system should display quality and production information through a centralized dashboard.
- **Digital Reporting:** The system should generate digital quality and production reports.
- **Historical Data Storage:** The system should store previous defect and production data for future analysis.
- **User and System Management:** Administrators should be able to manage users, system settings, defect categories, and quality thresholds.

6.4 Non-Functional Requirements

Requirement	Expected Quality
Performance	The system should process information quickly enough to support production decisions.
Reliability	Quality and production data should be stored consistently.
Scalability	The platform should support additional machines and production lines.
Security	Only authorized users should access sensitive production information.
Usability	The dashboard should be understandable to users with different technical backgrounds.
Maintainability	Individual components should be updateable without redesigning the entire platform.
Availability	The system should be available during required production operations.
Accuracy	The AI model should be evaluated using appropriate testing data and performance measures.

6.5 Software Engineering Principles

The system should be designed using modularity, separation of concerns, scalability, maintainability, reusability, security, and data integrity. For example, the defect detection component should be independent from the reporting component. This makes it possible to improve the machine-learning model without unnecessarily changing the entire application.

7. Development Methodology :

7.1 Recommended Methodology: Agile Scrum

Agile Scrum is suitable for this project because the requirements and technical details may evolve during development. The project includes both traditional software development and AI model development. The performance of the AI component may depend on the availability and quality of training data, and user feedback may lead to changes in the dashboard and reporting features.

Scrum allows the platform to be developed incrementally. A basic version of the system can be built first, followed by improvements to the defect detection model, monitoring functions, analytics, and optimization features.

7.2 Proposed Sprint Plan

Sprint	Main Work	Expected Result
1	Requirements, stakeholder analysis, architecture, and database planning	Approved system design
2	Authentication, user roles, database, and core services	Working platform foundation
3	Dataset preparation, preprocessing, and initial AI model	Initial defect detection capability
4	Production monitoring and dashboard	Working monitoring interface
5	Alerts, trends, and root-cause analysis	Integrated analytical functions
6	Reports and optimization recommendations	Decision-support features
7	Integration, testing, evaluation, and refinement	Tested prototype

7.3 Reasons for Choosing Agile Scrum

- Requirements can be refined through regular feedback.
- AI models can be improved iteratively as additional data becomes available.
- Users can review early versions of the dashboard and suggest improvements.
- Technical risks can be identified earlier than in a single final delivery.
- Working features can be delivered in stages.

8. Expected Outcomes and Benefits :

If implemented successfully, the proposed platform can improve both quality control and production decision-making. The expected benefits are:

- Reduced manual inspection time and dependence on continuous visual checking.
- Earlier identification of defects and reduced material wastage.
- More consistent classification of supported defect types.
- Improved visibility of machine performance and production bottlenecks.
- Faster investigation of recurring defects.
- Better support for preventive maintenance and reduced unplanned downtime.
- Improved machine utilization and production scheduling.
- Centralized and traceable quality and productivity information.

9. Risks, Limitations and Challenges :

Risk / Limitation	Possible Effect	Possible Response
Insufficient training data	Poor performance on unseen fabric types or defects	Expand and improve the dataset.
False predictions	Some defects may be missed or incorrectly identified	Use confidence thresholds and human verification.
Lighting and camera variation	Image quality may affect detection	Standardize installation and preprocessing.
Different machine	Integration may be	Use an integration layer and

interfaces	technically difficult	standard data formats.
User resistance	Employees may be reluctant to adopt the system	Provide training and emphasize decision support.
Cybersecurity threats	Production data could be accessed improperly	Use authentication, authorization, and secure communication.
Implementation cost	Equipment and integration require investment	Use phased implementation and prioritize high-value areas.

10. Future Enhancements :

- Predictive maintenance using historical machine sensor data.
- Edge computing for faster image processing close to the production line.
- Energy monitoring and energy-use optimization.
- Integration with ERP, inventory, order management, and supply-chain systems.
- Cloud-based monitoring across multiple factories.
- Digital twin technology for production simulation.
- Additional deep-learning models for more complex defect types.
- Human-in-the-loop learning to improve the AI model using verified inspector feedback.

11. Conclusion :

The problem analyzed in this report is not limited to the difficulty of identifying fabric defects. The broader issue is the lack of connection between quality inspection, machine performance, production conditions, and management decision-making. Manual inspection and disconnected records make it difficult to detect problems early and understand why they occur.

The proposed Smart Textile Manufacturing Quality Inspection and Production Optimization Platform addresses this issue by combining automated defect detection, production monitoring, centralized data management, real-time alerts, root-cause analysis, digital reporting, and optimization support.

The solution is designed to assist human decision-makers rather than assume that automation alone can solve every manufacturing problem. Its success will depend on the quality of the training data, the reliability of the sensors and cameras, the accuracy of the software, and the willingness of users to adopt the system. Nevertheless, the platform provides a practical foundation for improving quality, reducing waste, and moving textile manufacturing toward a more data-driven and proactive operating model.

12. References :

The final submission should use the referencing style required by the institution. Relevant reference areas for this case study include:

- Computer vision and machine learning for textile defect detection.
- Industrial Internet of Things and smart manufacturing.
- Machine-learning approaches to predictive maintenance and root-cause analysis.
- Production scheduling and optimization.
- Agile Scrum and software development life-cycle methodologies.
- Software engineering principles relating to modularity, scalability, maintainability, security, and data integrity.

Appendix A: Summary of the Proposed Platform :

System Component	Main Purpose
Cameras	Capture fabric images for inspection.
Machine Learning	Detect and classify supported defects.
IoT / Machine Data	Provide information about production conditions.

Central Database	Store quality and production records.
Analytics	Identify trends and possible relationships.
Dashboard	Display current performance and quality information.
Alerts	Notify users of important conditions.
Reports	Provide digital records for review and decision-making.
Optimization Support	Recommend possible production improvements.

